PHYS 7502 – Homework 5: Problems 6–9

PDFs: Sum Rules, Isospin Symmetry, Structure Function Ratios, and DGLAP Evolution

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References: Thomson, Modern Particle Physics*, Ch. 8 & 12; Griffiths,* Introduction to Elementary Particles*, Ch. 8; Halzen & Martin,* Quarks and Leptons*, Chs. 9–10; Mann,* An Introduction to Particle Physics*, Ch. 17.*

Background you need before touching these problems

All four problems live in the world of *deep inelastic scattering* (DIS) and parton distribution functions (PDFs). Before jumping into the solutions, let me lay out the five key ideas you need.

Idea 1: What is a PDF?

A **parton distribution function** $q(x)$ is a probability density: $q(x)\,dx$ is the average number of partons of flavor q inside the proton carrying a momentum fraction between x and $x + dx$. For example, $u(x)\,dx$ counts the number of up quarks (both valence *and* sea) with momentum fraction in $[x, x + dx]$.

Key point: $u(x)$ is *not* the probability of finding "the up quark" – it is a *number density*. Integrating $u(x)$ over all x gives the *average number* of up quarks in the proton (valence + sea), not 1.

Idea 2: Valence vs. sea decomposition

We split each quark PDF into a **valence** part (the "permanent" quarks that give the proton its quantum numbers) and a **sea** part (virtual $q\bar{q}$ pairs from gluon splitting):

$$u(x) = u_V(x) + u_S(x), \qquad d(x) = d_V(x) + d_S(x).$$

Antiquarks come *only* from the sea: $\bar{u}(x) = u_S(x)$, $\bar{d}(x) = d_S(x)$.

Because the proton is uud, there are two valence up quarks and one valence down:

$$\boxed{\int_0^1 u_V(x)\,dx = 2, \qquad \int_0^1 d_V(x)\,dx = 1.} \tag{1}$$

Since $q\bar{q}$ are produced in *pairs*, the sea quark and the corresponding antiquark have the same PDF: $u_S(x) = \bar{u}(x)$, $d_S(x) = \bar{d}(x)$. And if we assume $m_u \approx m_d$, the sea is approximately flavor-symmetric: $u_S(x) \approx d_S(x) \approx \bar{u}(x) \approx \bar{d}(x) \equiv S(x)$.

Idea 3: Structure functions from PDFs

In the quark–parton model, the proton and neutron electromagnetic structure functions are charge-squared-weighted sums of PDFs:

$$F_2^{ep}(x) = x\Big[\big(\tfrac{2}{3}\big)^2\big(u(x)+\bar{u}(x)\big) + \big(\tfrac{1}{3}\big)^2\big(d(x)+\bar{d}(x)\big)\Big] = x\big[\tfrac{4}{9}\,u(x) + \tfrac{1}{9}\,d(x) + \tfrac{4}{9}\,\bar{u}(x) + \tfrac{1}{9}\,\bar{d}(x)\big]\,, \tag{2}$$

$$F_2^{en}(x) = x\big[\tfrac{4}{9}\,u^n(x) + \tfrac{1}{9}\,d^n(x) + \tfrac{4}{9}\,\bar{u}^n(x) + \tfrac{1}{9}\,\bar{d}^n(x)\big]\,. \tag{3}$$

The charges $Q_u = +\frac{2}{3}$ and $Q_d = -\frac{1}{3}$ enter as Q^2, giving the weights $\frac{4}{9}$ and $\frac{1}{9}$.

Idea 4: Isospin symmetry – relating neutron to proton

The proton is uud; the neutron is ddu. Under the strong interaction, u and d are approximately interchangeable (isospin symmetry). So the *distribution of the up quark in the neutron* is the same as the *distribution of the down quark in the proton*, and vice versa:

$$\boxed{u^n(x) = d^p(x) \equiv d(x), \qquad d^n(x) = u^p(x) \equiv u(x),} \tag{4}$$

and likewise $\bar{u}^n(x) = \bar{d}(x)$, $\bar{d}^n(x) = \bar{u}(x)$. This is the single most important trick in this problem set. It lets us write *everything* in terms of proton PDFs.

Idea 5: Bjorken scaling and its violations (DGLAP)

At leading order ("naive parton model"), the structure functions depend only on x, not on Q^2 – this is **Bjorken scaling**. It holds because the virtual photon scatters off a *pointlike* quark.
In reality, QCD corrections break exact scaling. At higher Q^2 (shorter wavelength), the photon resolves finer structure: gluon radiation $q \to qg$ and pair production $g \to q\bar{q}$ redistribute momentum from high-x partons to low-x partons. The result: F_2 *increases* at low x and *decreases* at high x as Q^2 grows. These are **scaling violations**.
The **DGLAP equations** (Dokshitzer–Gribov–Lipatov–Altarelli–Parisi) are the differential equations that predict *exactly how* the PDFs change with Q^2: if you know $q(x, Q_0^2)$ at one scale, DGLAP evolves it to any other scale Q^2.

Problem 6 **(Thomson 8.5)**

Problem 6 – Restated

What is the expected value of

$$\int_0^1 \big[u(x) - \bar{u}(x)\big]\,dx$$

for the proton? (The clock icon on the original problem indicates this should be quick – but justify your answer.)

Deliverables

- State the valence/sea decomposition of $u(x)$ and $\bar{u}(x)$.
- Show that the integral reduces to the valence normalization.
- Give the numerical answer with one-line justification.

What this integral is asking physically

The integrand $u(x) - \bar{u}(x)$ is the *net number density of up quarks* at momentum fraction x. At each x, sea up quarks and sea anti-up quarks cancel perfectly (they are produced in $q\bar{q}$ pairs), so the difference picks out only the valence content.

Solution

We use the decomposition $u(x) = u_V(x) + u_S(x)$ and $\bar{u}(x) = u_S(x)$:

$$\int_0^1 \big[u(x) - \bar{u}(x)\big]\, dx = \int_0^1 \big[u_V(x) + u_S(x) - u_S(x)\big]\, dx = \int_0^1 u_V(x)\, dx. \tag{5}$$

But the proton contains exactly 2 valence up quarks, so by the normalization condition (1):

Key Result

$$\boxed{\int_0^1 \big[u(x) - \bar{u}(x)\big]\, dx \;=\; 2.}$$

This is the **up-quark number sum rule**: the net number of up quarks in the proton is 2, regardless of how many sea pairs are present.

Why sea quarks don't affect this

No matter how many $u\bar{u}$ pairs the gluons produce, each pair adds $+1$ to $\int u(x)\, dx$ and $+1$ to $\int \bar{u}(x)\, dx$, so their difference is always zero. The integral $\int [u - \bar{u}]\, dx$ is "blind" to the sea: it counts only the non-perturbative valence content that gives the proton its quantum numbers ($B = 1$, $I_3 = +\frac{1}{2}$, $Q = +1$). These are exact QCD sum rules, valid to all orders in α_s.

Companion sum rules (Griffiths 8.16)

By identical reasoning:

$$\int_0^1 \big[d(x) - \bar{d}(x)\big]\, dx = 1, \qquad \int_0^1 \big[s(x) - \bar{s}(x)\big]\, dx = 0.$$

The strange-quark integral is zero because there are no valence strange quarks in the proton. Adding all three gives the **Gross–Llewellyn-Smith sum rule**: $\int_0^1 [u_V(x) + d_V(x)]\, dx = 3$ (the baryon number of the proton).

Problem 7 (Griffiths 1st ed., Problem 8.15)

Problem 7 – Restated

Given the experimentally measured structure function integrals

$$\int_0^1 F_2^{\text{neutron}}(x)\,dx = 0.12, \qquad \int_0^1 F_2^{\text{proton}}(x)\,dx = 0.18,$$

use these to confirm Griffiths' equation 8.57:

$$\int_0^1 x\,u(x)\,dx \;=\; 2\int_0^1 x\,d(x)\,dx.$$

Hint (from Griffiths): How do you suppose $u^n(x)$ and $d^n(x)$ are related to the corresponding functions for the proton?

Deliverables

- Use isospin symmetry to write F_2^{en} in terms of proton PDFs.
- Define momentum fractions f_u and f_d.
- Solve the two-equation system from the two experimental integrals.
- Show that the solution implies $f_u = 2f_d$, confirming Eq. 8.57.

Background: what Griffiths Eq. 8.57 says

Griffiths defines $u(x)$ and $d(x)$ as the *full* (valence + sea + antiquark) quark distribution functions. In his notation (which at this level ignores the separate $\bar{u}$, $\bar{d}$ contributions – or equivalently folds them in), $F_2^p(x) = x\left[\frac{4}{9}\,u(x) + \frac{1}{9}\,d(x)\right]$.

Eq. 8.57 states:

$$\int_0^1 x\,u(x)\,dx = 2\int_0^1 x\,d(x)\,dx. \qquad \text{(Griff 8.57)}$$

Physically, this says: the total momentum fraction carried by up-type quarks is twice that carried by down-type quarks. Since the proton has 2 u's and 1 d, and if on average each quark carries the same momentum, this is exactly what you'd expect.

Step 1: Write structure functions using isospin symmetry

This is where the Griffiths hint comes in. The neutron is the proton with $u \leftrightarrow d$ swapped (Eq. 4):

$$u^n(x) = d(x), \qquad d^n(x) = u(x).$$

So the neutron structure function, in terms of proton PDFs, is:

$$F_2^n(x) = x\left[\tfrac{4}{9}\,d(x) + \tfrac{1}{9}\,u(x)\right]. \qquad (6)$$

(Just swap $u \leftrightarrow d$ everywhere in F_2^p.)

Step 2: Integrate both structure functions

Define the momentum fractions:

$$f_u \equiv \int_0^1 x\,u(x)\,dx, \qquad f_d \equiv \int_0^1 x\,d(x)\,dx. \tag{7}$$

These are the fractions of the proton's total momentum carried by all up-type quarks (valence + sea + antiquarks) and all down-type quarks respectively.

Integrating:

$$\int_0^1 F_2^p(x)\,dx = \tfrac{4}{9}\,f_u + \tfrac{1}{9}\,f_d = 0.18, \tag{8}$$

$$\int_0^1 F_2^n(x)\,dx = \tfrac{4}{9}\,f_d + \tfrac{1}{9}\,f_u = 0.12. \tag{9}$$

Step 3: Solve the system

We have two equations in two unknowns. From Eq. (8):

$$4f_u + f_d = 9 \times 0.18 = 1.62.$$

From Eq. (9):

$$f_u + 4f_d = 9 \times 0.12 = 1.08.$$

Multiply the second by 4 and subtract the first:

$$4(f_u + 4f_d) - (4f_u + f_d) = 4(1.08) - 1.62 \quad \Longrightarrow \quad 15\,f_d = 2.70 \quad \Longrightarrow \quad f_d = 0.18.$$

Substituting back: $4f_u + 0.18 = 1.62 \;\Rightarrow\; f_u = 0.36$.

Key Result

$$\boxed{f_u = 0.36, \qquad f_d = 0.18, \qquad f_u = 2\,f_d.}$$

This confirms Griffiths Eq. 8.57: $\int_0^1 x\,u(x)\,dx = 2\int_0^1 x\,d(x)\,dx$.

What this teaches us

- The proton has twice as many up quarks as down quarks. If each quark on average carries the same momentum, the total up-quark momentum fraction should be double the down-quark fraction – and it is.
- $f_u + f_d = 0.54$: quarks carry only 54% of the proton's total momentum. The remaining 46% is carried by **gluons**, which are electrically neutral and invisible to the virtual photon in DIS. This was one of the strongest early pieces of evidence for gluons.
- The fact that $f_u = 2f_d$ *exactly* is a consequence of isospin symmetry and the specific numerical values 0.18/0.12. In reality there are small corrections from the sea and from isospin-breaking effects ($m_u \neq m_d$), but the leading-order picture holds beautifully.

Problem 8 **(Halzen & Martin Fig. 9.6)**

Problem 8 – Restated

From the Halzen & Martin reading, discuss the theoretical understanding of the plot of $F_2^n(x)/F_2^p(x)$ versus x (H&M Figure 9.6; equivalent to Thomson Figure 8.12). Specifically: what does the quark–parton model predict for this ratio at the **left-most side** ($x \to 0$) and the **right-most side** ($x \to 1$), and why?

Deliverables

- Write the general expression for F_2^n/F_2^p in terms of valence and sea PDFs.
- Evaluate the $x \to 0$ limit (sea-dominated).
- Evaluate the $x \to 1$ limit (valence-dominated), first under the naive assumption $u_V = 2d_V$, then using the experimental data.
- Explain the physics behind the $x \to 1$ suppression of d_V/u_V.

The general ratio

Using the valence + sea decomposition (Thomson Eq. 8.30):

$$\frac{F_2^n(x)}{F_2^p(x)} = \frac{4\,d_V(x) + u_V(x) + 10\,S(x)}{4\,u_V(x) + d_V(x) + 10\,S(x)}, \tag{10}$$

where we used the flavor-symmetric sea approximation $u_S = d_S = \bar{u} = \bar{d} \equiv S(x)$.

Left side: $x \to 0$ (sea-dominated)

At very low x, the gluon-produced sea quarks dominate. Sea quarks are produced via $g \to q\bar{q}$, and since $m_u \approx m_d$, the process creates $u\bar{u}$ and $d\bar{d}$ pairs in roughly equal proportions. So as $x \to 0$, $S(x) \gg u_V(x), d_V(x)$.

In Eq. (10), the valence terms become negligible relative to $10\,S(x)$ in both numerator and denominator:

$$\frac{F_2^n}{F_2^p} \xrightarrow{x\to 0} \frac{10\,S(x)}{10\,S(x)} = 1. \tag{11}$$

Key Result

At low x, the proton and neutron look identical because both are dominated by the same flavor-symmetric quark–antiquark sea: $F_2^n/F_2^p \to 1$.

Right side: $x \to 1$ (valence-dominated)

At high x, one valence quark is carrying most of the proton's momentum. Sea quarks are suppressed (they originate from gluons, which are soft due to the $1/q^2$ propagator). So $S(x) \to 0$ and Eq. (10) becomes:

$$\frac{F_2^n}{F_2^p} \xrightarrow{x\to 1} \frac{4\,d_V(x) + u_V(x)}{4\,u_V(x) + d_V(x)}. \tag{12}$$

Naive guess: $u_V(x) = 2\,d_V(x)$. If we assume the two valence up quarks and one valence down quark have the same x-shape (just different normalization), then $u_V = 2d_V$ everywhere, and:

$$\frac{F_2^n}{F_2^p} \to \frac{4(1)+2}{4(2)+1} = \frac{6}{9} = \frac{2}{3} \approx 0.67.$$

What the data actually show: ratio $\to 1/4$. The experimental data (H&M Fig. 9.6 / Thomson Fig. 8.12) show the ratio falling well *below* 2/3, approaching $\sim 1/4$ as $x \to 1$. This implies $d_V(x)/u_V(x) \to 0$ as $x \to 1$.

Plugging $d_V \to 0$ into Eq. (12):

$$\frac{F_2^n}{F_2^p} \to \frac{0 + u_V}{4u_V + 0} = \frac{1}{4}.$$

Key Result

$$\frac{F_2^n(x)}{F_2^p(x)} \xrightarrow{x\to 1} \frac{1}{4} \qquad \text{(experimentally observed).}$$

This implies the down-quark valence PDF is suppressed relative to the up-quark at high x: $d_V(x)/u_V(x) \to 0$ as $x \to 1$.

Why is d_V suppressed at high x? – The Pauli exclusion argument

When $x \to 1$, one quark carries almost all the momentum and the other two valence quarks are forced into a very low-momentum state. Now consider the possibilities:

1. **A u quark is at high x:** The remaining two quarks are one u and one d, which are distinguishable. There is no Pauli exclusion constraint on this low-momentum (u, d) pair – they can happily occupy the same low-momentum state. This configuration is *allowed*.
2. **The d quark is at high x:** The remaining two quarks are both u's, and they must share a very low-momentum state. But the Pauli exclusion principle forbids two identical fermions from occupying the same quantum state (same flavor, same color, same spin, same momentum). This configuration is *suppressed*.

So at high x, the proton strongly prefers to have a u quark carrying the bulk of the momentum, not the d. This drives $d_V(x)/u_V(x) \to 0$ and $F_2^n/F_2^p \to 1/4$.

Note that the $\frac{1}{4}$ value is the ratio of the squared charges: $Q_u^2/Q_d^2 = (2/3)^2/(1/3)^2 = 4$. When only up quarks remain at high x, the neutron (which has that u as a minority) sees it weighted by $Q_d^2 = \frac{1}{9}$ while the proton sees it weighted by $Q_u^2 = \frac{4}{9}$, hence the ratio $\frac{1/9}{4/9} = \frac{1}{4}$.

Problem 9 **(DGLAP evolution and scaling violations)**

Problem 9 – Restated

This problem has two parts:

Part (a): Explain what DGLAP evolution of PDFs is useful for.

Part (b): Describe the relation between DGLAP and H&M Figure 10.10 (or similar plots from slides showing $F_2(x, Q^2)$ vs. Q^2 at different x). Also explain in $\sim$2 sentences: what are the "scaling violations" on the figure, and what do they represent physically?

(The answer to part (a) is primarily lecture material but can also be deduced from the H&M reading.)

Deliverables

- Define Bjorken scaling and explain why it is expected at leading order.
- Explain what scaling *violations* are.
- State the physical mechanism (QCD radiation) that causes them.
- State what the DGLAP equations do (evolve PDFs in Q^2).
- Explain why DGLAP is useful in practice (predictions at new Q^2 scales, e.g. LHC from HERA data).
- Interpret the log-Q^2 plot: F_2 rises at low x, falls at high x, is approximately flat at medium x.

Part (a): What is DGLAP evolution useful for?

The problem DGLAP solves

In the naive parton model, the PDFs $q(x)$ depend only on x, not on Q^2 – this is Bjorken scaling. But we know from QCD that quarks can radiate gluons ($q \to qg$) and gluons can split into quark pairs ($g \to q\bar{q}$). These processes *redistribute momentum* among the partons, and the extent to which they do so depends on Q^2 because higher Q^2 probes shorter distances and "resolves" more radiation.
So the PDFs really are functions of *two* variables: $q(x, Q^2)$.
The **DGLAP equations** are a system of integro-differential equations that predict how the PDFs change with Q^2:

$$\frac{\partial q_i(x,Q^2)}{\partial \ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi} \sum_j \int_x^1 \frac{dz}{z} P_{ij}(z)\, q_j\left(\frac{x}{z}, Q^2\right), \tag{13}$$

where $P_{ij}(z)$ are the **splitting functions** – calculable in perturbative QCD – that describe the probability for parton j to "split" into parton i carrying a fraction z of the parent's momentum.

Why DGLAP is essential in practice

1. **Prediction at new energy scales.** PDFs are measured at one Q^2 (say, fixed-target DIS at $Q^2 \sim 10\,\text{GeV}^2$). To predict cross sections at the LHC ($Q^2 \sim 10^4\,\text{GeV}^2$), you need to know what the PDFs look like at that much higher scale. DGLAP evolves them from the measurement scale to the prediction scale.
2. **Testing QCD.** The fact that the Q^2 evolution predicted by DGLAP (using only the QCD coupling α_s and the perturbatively calculated splitting functions) matches the observed scaling violations is one of the most powerful quantitative tests of QCD. The agreement spans four orders of magnitude in Q^2 and two orders in x.
3. **Global PDF fits.** Modern PDF sets (NNPDF, CT, MSHT) are determined by fitting experimental data at many different Q^2 values simultaneously, using DGLAP to connect them. Without DGLAP, measurements at different Q^2 could not be combined.

Part (b): What do the scaling violations look like and represent?

The H&M Figure 10.10 (and similar HERA/SLAC/Thomson plots) show $F_2(x, Q^2)$ plotted against $\ln Q^2$ at several fixed values of x. If Bjorken scaling were exact, each curve would be a horizontal line. Instead, three distinct behaviors are observed:

The three regimes of scaling violation

1. **At high x ($x \gtrsim 0.3$): F_2 decreases with increasing Q^2.**
   Physical reason: a high-x quark is likely to have radiated a gluon ($q \to qg$), losing some momentum. The higher Q^2 is, the more of this radiation you resolve, so the effective PDF at high x is depleted. Momentum is "drained" from high x to low x.
2. **At low x ($x \lesssim 0.1$): F_2 increases with increasing Q^2.**
   Physical reason: this is where the radiated gluons and their daughter $q\bar{q}$ pairs end up. At higher Q^2, you resolve more $g \to q\bar{q}$ splittings, so there are more low-x sea quarks visible to the photon. Momentum that was "drained" from high x accumulates here.
3. **At intermediate x ($x \approx 0.15$–0.25): F_2 is approximately constant.**
   This is the pivot point where the inflow from above and outflow to below balance. The structure function is roughly Q^2-independent here, and approximate Bjorken scaling holds.

Two-sentence summary of scaling violations

Scaling violations are the observed Q^2-dependence of the structure functions $F_2(x, Q^2)$: F_2 rises at low x and falls at high x as Q^2 increases. Physically, they represent the QCD processes $q \to qg$ and $g \to q\bar{q}$, which redistribute momentum from high-x partons to low-x partons as the virtual photon probes shorter distances at higher Q^2.

Connection between the figure and DGLAP

The slopes $\partial F_2 / \partial \ln Q^2$ that you see on the H&M plot are *precisely what DGLAP predicts*. At each x, DGLAP sums over all the ways a parton at $x' > x$ could have radiated down to x (via the splitting functions P_{qq}, P_{qg}, P_{gq}, P_{gg}), weighted by $\alpha_s(Q^2)$. The fact that the *measured* slopes agree with the DGLAP prediction over a huge range of x and Q^2 is among the strongest evidence that QCD is the correct theory of the strong interaction.
The practical upshot: measure PDFs at HERA ($Q^2 \sim 10$–$10^4\,\text{GeV}^2$), evolve them with DGLAP to LHC energies ($Q^2 \sim M_H^2 \sim 1.6 \times 10^4\,\text{GeV}^2$), and predict the Higgs production cross section. This is exactly how the Higgs boson cross section was calculated before its discovery.

Summary of Results

All four answers at a glance

- **Problem 6 (Thomson 8.5):** $\int_0^1 [u(x) - \bar{u}(x)]\,dx = 2$ (the number of valence up quarks).
- **Problem 7 (Griffiths 8.15):** $f_u = 0.36$, $f_d = 0.18$, so $f_u = 2f_d$, confirming Eq. 8.57. Quarks carry only 54% of the proton momentum; gluons carry the rest.
- **Problem 8 (H&M Fig. 9.6):** $F_2^n / F_2^p \to 1$ as $x \to 0$ (flavor-symmetric sea dominates) and $\to 1/4$ as $x \to 1$ ($d_V/u_V \to 0$ due to Pauli exclusion).
- **Problem 9 (DGLAP):** (a) DGLAP evolves PDFs from one Q^2 to another, enabling predictions at new energy scales and providing a quantitative test of QCD. (b) Scaling violations $= Q^2$-dependence of F_2; physically, QCD radiation ($q \to qg$, $g \to q\bar{q}$) moves momentum from high x to low x as Q^2 increases.